Number Relationships - Mark scheme

Number · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

Question 1

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2025 · Question 13

Q	Working	Answer	Mark	Notes
	eg $\frac{12}{28}$ and $\frac{7}{28}$ **or** $\frac{12n}{28n}$ and $\frac{7n}{28n}$ **or** $4 \times 3\ (=12)$ **and** $(1\times)\ 7$ **and** $4 \times 7\ (=28)$ **or** $\frac{3\times4}{4\times7}$ and $\frac{(1\times)7}{4\times7}$		2	M1 for finding a common denominator with at least one fraction correct **or** for calculations that would lead to 12 **and** 7 **and** 28
	$\frac{12}{28}+\frac{7}{28}=\frac{19}{28}$ **or** $\frac{12n}{28n}+\frac{7n}{28n}=\frac{19n}{28n}=\frac{19}{28}$ **or** $12+7=19$ **and** $\frac{19}{28}$ **or** $4\times3+(1\times)\ 7\ (=19)$ **and** $4\times7\ (=28)$ **and** $\frac{19}{28}$ **or** $\frac{3\times4+(1\times)7}{4\times7}=\frac{19}{28}$ **or** $\frac{3\times4}{4\times7}+\frac{(1\times)7}{4\times7}=\frac{19}{28}$ *Working required*	Shown		A1 dep on M1, for a complete correct method leading to $\frac{19}{28}$
				Total 2 marks

Question 2

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · January 2023 · Question 13

13	eg 10 000x − 3818.18… 100x = 38.18… **or** 1000x = 381.818… 10x = 3.818… **or** 100x = 38.1818… x = 0.3818… oe		2	M1 for **selecting** 2 correct recurring decimals that when subtracted give a whole number or terminating decimal (37.8 or 378 or 3780 etc) eg 10 000x = 3818.18... and 100x = 38.1818.... **or** 1000x = 381.818… and 10x = 3.81818… **or** 100x = 38.1818… and x = 0.381818… with intention to subtract. (if recurring dots not shown then showing at least **one** of the numbers to at least 5 sf) **or** $0.38+0.00\dot{1}\dot{8}$ **and** eg $100x=0.1818...$, $10000x=18.1818...$ with intention to subtract.
	eg 10 000x − 100x = 3818.18... − 38.1818...= 3780 (9900x = 3780) **and** $\frac{3780}{9900}=\frac{21}{55}$ **or** eg 1000x − 10x = 381.818... − 3.81818...= 378 (990x = 378) **and** $\frac{378}{990}=\frac{21}{55}$ **or** eg 100x − x = 38.1818... − 0.381818...= 37.8 (99x = 37.8) **and** $\frac{37.8}{99}=\frac{21}{55}$ **or** eg 10 000x − 100x = 18.1818... − 0.181818...= 18 **and** $0.38+\frac{18}{9900}=\frac{38\times99+18}{9900}=\frac{3780}{9900}=\frac{21}{55}$ oe	shown		A1 for completion to $\frac{21}{55}$ dep on M1 *(NB: this is a "use algebra to show that…" question, so we need to see algebra as well as seeing all the stages of working to award full marks)*
				Total 2 marks

Question 3

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2024 · Question 15

15		eg $(10\,000x =)\ 7636.36\ldots$ $\underline{\quad (100x =)\quad 76.36\ldots}$ **or** $(1000x =)\ 763.63\ldots$ $\underline{\quad (10x =)\quad 7.63\ldots}$ **or** $(100x =)\ 76.363\ldots$ $\underline{\quad (x =)\quad 0.763\ldots}$ oe		2	M1	For 2 recurring decimals that when subtracted give a whole number or terminating decimal with intention to subtract. (ie give 75.6 or 756 or 7560 etc) eg $(10\,000x =)\ 7636.36...$ and $(100x =)\ 76.36....$ **or** $(1000x =)\ 763.63\ldots$ and $(10x =)\ 7.63\ldots$ **or** $(100x =)\ 76.363\ldots$ and $(x =)\ 0.763\ldots$ with intention to subtract. (if recurring not shown then showing at least **one** of the numbers to at least 5sf) or $\frac{7}{10}+1000x(63.6\dot{3})-10x(0.6\dot{3})$
		eg $10\,000x - 100x = 7636.36... - 76.36... = 7560$ **and** $\frac{7560}{9900}=\frac{42}{55}$ **or** $1000x - 10x = 763.63... - 7.63... = 756$ $(990x = 756)$ **and** $\frac{756}{990}=\frac{42}{55}$ **or** $100x - x = 76.363... - 0.763... = 75.6$ **and** $\frac{75.6}{99}=\frac{42}{55}$ $(99x = 75.6)$ **or** $\frac{7}{10}+\frac{63}{990}=\frac{7\times99+63}{990}=\frac{42}{55}$	shown		A1	for completion to $\frac{42}{55}$ dep on M1 and must use algebra for this final mark to be awarded [allow for instance $99x = 75.6$ and then $\frac{756}{990}=\frac{42}{55}$] **No algebra used gets a maximum of 1**
		Working required				**Total 2 marks**

Question 4

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2025 · Question 16

16	eg $(100x =)\ 37.878\ldots$ $\underline{\quad (x =)\ 0.378\ldots}$ **or** $(1000x =)\ 378.78\ldots$ $\underline{\quad (10x =)\quad 3.78\ldots}$ **or** $(100000x =)\ 37878.78\ldots$ $\underline{\quad (1000x =)\quad 378.78\ldots}$		2	M1 for 2 **correct** recurring decimals that when subtracted should give a whole number or terminating decimal with intention to subtract. (ie may give 37.5 or 375 or 37500 or 78 etc) eg $(1000x =)\ 37.878....$ and $(x =)\ 0.378....$ **or** $(1000x =)\ 378.78....$ and $(10x =)\ 3.78....$ **or** $(100000x =)\ 37878.78....$ and $(1000x =)\ 378.78....$ **or** $0.3+0.07\dot{8}$ **and** $(1000y =)\ 78.787...$ and $(10y =)\ 0.78....$ with intention to subtract. x is not required to award this mark (if recurring dots not shown in both numbers then showing at least **one** of the numbers to at least **5sf**) **NB** Accept bar notation for dot notation to indicate recurring decimals
	eg $100x - x = 37.878\ldots - 0.378\ldots = 37.5$ **and** $\frac{37.5}{99}=\frac{25}{66}$ **or** $1000x - 10x = 378.78... - 3.78... = 375$ **and** $\frac{375}{990}=\frac{25}{66}$ **or** $100000x - 1000x = 37878.78... - 378.78... = 37500$ **and** $\frac{37500}{99000}=\frac{25}{66}$ **or** $1000y - 10y = 78.787... - 0.7878... = 78$ **and** $0.3+\frac{78}{990}=\frac{0.3\times990+78}{990}=\frac{375}{990}=\frac{25}{66}$ oe	Shown		A1 for completion to $\frac{25}{66}$ dep on M1and must use algebra for this final mark to be awarded **No algebra used gets a maximum of 1 mark**
	Working required			**Total 2 marks**

Question 5

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2025 · Question 16

16	$(10000x =)\ 6121.21\ldots$ eg $\underline{(100x =)\ \ 61.21\ldots}$ $(1000x =)\ 612.12\ldots$ **or** $\underline{(10x =)\ \ 6.12\ldots}$ $(100x =)\ 61.212\ldots$ **or** $\underline{(x =)\ \ 0.612\ldots}$		2	M1	for 2 recurring decimals that when subtracted give a whole number or terminating decimal with intention to subtract. (ie give 60.6 or 606 or 6060 etc) eg $(1000x =)\ 612.12\ldots$ and $(10x =)\ 6.12\ldots$ **or** $(100\,000x =)\ 61\,212.12\ldots$ and $(1000x =)\ 612\ldots$ **or** $(100x =)\ 61.212\ldots$ and $(x =)\ 0.612\ldots$ with intention to subtract x is not required to award this mark (if recurring dots not shown in **both** numbers then showing at least **one** of the numbers to at least 5sf) or $\frac{6}{10}+1000x(12.12)-10x(0.12)$
	eg $10000x - 100x = 6121.21\ldots - 61.21\ldots = 6060$ $(9900x = 6060)$ **and** $\frac{6060}{9900}=\frac{101}{165}$ **or** $1000x - 10x = 612.12\ldots - 6.12\ldots = 606$ $(990x = 606)$ **and** $\frac{606}{990}=\frac{101}{165}$ **or** $100x - x = 61.212\ldots - 0.612\ldots\ 60.6$ $(99x = 60.6)$ **and** $\frac{60.6}{99}=\frac{101}{165}$ oe **OR** $0.6+\ldots$ and $(1000x - 100x = 990x = 12)$ **and** $0.6+\frac{12}{990}=\frac{0.6\times990+12}{990}=\frac{101}{165}$ oe	shown		A1	for completion to $\frac{101}{165}$ dep on M1 and must use algebra for this final mark to be awarded [allow for instance $99x = 60.6$ and then $\frac{606}{990}=\frac{101}{165}$] **No algebra used gets a maximum of 1 mark**
	Working required				**Total 2 marks**

Question 6

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2023 · Question 16

16		eg $10\,000x = 1767.67\ldots$ $\underline{\ \ 100x = \ \ 17.67\ldots}$ **or** $1000x = 176.76\ldots$ $\underline{\ \ 10x = \ \ 1.76\ldots}$ **or** $100x = 17.676\ldots$ $\underline{\ \ x = \ \ 0.176\ldots}$ oe		2	M1	for 2 recurring decimals that when subtracted give a whole number or terminating decimal (17.5 or 175 or 1750 etc) eg $10\,000x = 1767.67\ldots$ and $100x = 17.676\ldots$ **or** $1000x = 176.76\ldots$ and $10x = 1.7676\ldots$ **or** $100x = 17.676\ldots$ and $x = 0.17676\ldots$ with intention to subtract. (if recurring dots not shown in both numbers then showing at least **one** of the numbers to at least 5sf) **or** $0.1+0.0\dot{7}\dot{6}$ **and** eg $100x = 7.6767\ldots$, $x = 0.07676\ldots$ with intention to subtract.
		eg $10\,000x - 100x = 1767.67\ldots - 17.67\ldots = 1750$ **and** $\frac{1750}{9900}=\frac{35}{198}$ **or** $1000x - 10x = 176.76\ldots - 1.76\ldots = 175$ **and** $\frac{175}{990}=\frac{35}{198}$ **or** $100x - x = 17.676\ldots - 0.176\ldots = 17.5$ **and** $\frac{17.5}{99}=\frac{35}{198}$ **or** eg $10x - x = 7.6767\ldots - 0.07676\ldots = 7.6$ **and** $0.1+\frac{7.6}{99}=\frac{0.1\times99+7.6}{99}=\frac{17.5}{99}=\frac{35}{198}$ oe	shown		A1	for completion to $\frac{35}{198}$ dep on M1
		Working required				**Total 2 marks**

A+ Maths Hub International Community Schools

Question 7

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2023 · Question 16

Question	Working	Answer	Mark	Notes
16	eg $1000x = 438.38\ldots$ $10x = 4.38\ldots$ **or** $100x = 43.838\ldots$ $x = 0.438\ldots$ oe		2	M1 For **selecting** 2 correct recurring decimals that when subtracted give a whole number or terminating decimal (43.4 or 434 etc) eg $1000x = 438.38...$ and $10x = 4.38....$ **or** $100x = 43.838\ldots$ and $x = 0.438\ldots$ with intention to subtract. (if recurring dots not shown then showing at least **one** of the numbers to at least 5sf) **or** $0.4+0.0\dot{3}\dot{8}$ **and** eg $1000x = 38.38...$ & $10x = 0.3838...$, with intention to subtract.
	eg $1000x - 10x = 438.38\ldots - 4.38\ldots = 434$ **and** $\frac{434}{990} = \frac{217}{495}$ **or** eg $100x - x = 43.838\ldots - 0.438\ldots = 43.4$ **and** $\frac{43.4}{99} = \frac{217}{495}$ **or** eg $1000x - 10x = 38.38\ldots - 0.3838 = 38$ and $0.4 + \frac{38}{990} = \frac{4\times99+38}{990} = \frac{434}{990} = \frac{217}{495}$ oe *working required*	Clearly shown		A1 For completion to $\frac{217}{495}$ dep on M1 and use of some algebra
				Total 2 marks

Question 8

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2026 · Question 17

17	eg $(100x =)\ 53.939\ldots$ $(x =)\ 0.539\ldots$ **or** $(1000x =)\ 539.39\ldots$ $(10x =)\ 5.39\ldots$ **or** $(100\,000x =)\ 539\,39.39\ldots$ $(1000x =)\ 539.39\ldots$ **or** $0.5+0.0\dot{3}\dot{9}$ **and** eg $(1000y =)\ 39.393\ldots$ $(10y =)\ 0.39\ldots$		2	M1 for 2 **correct** recurring decimals that when subtracted should give a whole number or terminating decimal **and** intention to subtract (ie may give 53.4 or 534 or 53 400 or 39 or 3900 etc) eg any of the following with intention to subtract $(100x =)$ 53.939... and $(x =)$ 0.539.... **or** $(1000x =)$ 539.39... and $(10x =)$ 5.39.... **or** $(100\,000x =)$ 53939.... and $(1000x =)$ 539.39.... **or** $0.5+0.0\dot{3}\dot{9}$ **and** $(1000y =)\ 39.393\ldots$ and $(10y =)\ 0.39\ldots$ x is not required to award this mark if recurring dots not shown in both numbers then showing at least **one** of the numbers to at least 5sf is sufficient NB Accept bar notation for dot notation to indicate recurring decimals
	$100x - x = 53.939\ldots - 0.539\ldots = 53.4$ **and** $\frac{53.4}{99} = \frac{89}{165}$ **or** $1000x - 10x = 539.39\ldots - 5.39\ldots = 534$ **and** $\frac{534}{990} = \frac{89}{165}$ **or** $100\,000x - 1000x = 53939 - 539 = 53\,400$ **and** $\frac{53\,400}{99\,000} = \frac{89}{165}$ **or** $1000y - 10y = 39.393\ldots - 0.39\ldots = 39$ **and** $0.5 + \frac{39}{990} = \frac{0.5\times990+39}{990} = \frac{534}{990} = \frac{89}{165}$	Shown		A1 dep on M1, for completion to $\frac{89}{165}$ must use algebra for this final mark to be awarded **No algebra used gets a maximum of 1 mark**
	Working required			**Total 2 marks**

A+ Maths Hub International Community Schools

Question 9

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2022 · Question 17

17	eg $x = 0.34545....$ **and** $100x = 34.545....$ with intention to subtract **OR** $10x = 3.4545....$ **and** $1000x = 345.45....$ with intention to subtract Must include algebra as the question asked for 'using algebra'		2	M1	for 2 recurring decimals (they must identify or show the pair they are using) that when subtracted give a whole number or terminating decimal eg $100x = 34.545....$ and $x = 0.34545....$ **OR** $1000x = 345.45....$ and $10x = 3.4545....$ with intention to subtract. (If recurring dots not shown then showing at least the digits 34545, i.e. 5sf for **one** of the numbers that they are using) **OR** $0.3 + 0.0454545....$ and Eg $10x = 0.454545...$ and $1000x = 45.4545..$
	eg $100x - x = 34.545.... - 0.34545....$ **and** $99x = 34.2$ **and** $\frac{34.2}{99} = \frac{19}{55}$ oe **OR** $1000x - 10x = 345.45.... - 3.4545....$ **and** $990x = 342$ **and** $\frac{342}{990} = \frac{19}{55}$ oe **OR** $0.3 + ...$ and $(1000x - 100x = 990x = 45)$ and $0.3 + \frac{45}{990} = \frac{3\times99+45}{990} = \frac{19}{55}$ oe	shown		A1	for completion to $\frac{19}{55}$
					Total 2 marks

Question 10

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2025 · Question 18

18	$(1000x =)$ 702.702..._ eg $(x =)$ 0.702... $(10000x =)$ 7027.02..._ **or** $(10x =)$ 7.02... $(100000x =)$ 70270.20..._ **or** $(100x =)$ 70.20...		2	M1	for 2 recurring decimals that when subtracted give a whole number or terminating decimal with intention to subtract. (ie give 702 or 7020 or 70200 etc) eg $(1000x =)$ 702.702... and $(x =)$ 0.702.... **or** $(10000x) = 7027.02....$ and $(10x) = 7.02....$ **or** $(100000x =)$ 70270.20.... and $(100x =)$ 70.20.... with intention to subtract x is not required to award this mark (if recurring dots not shown in both numbers then showing at least **one** of the numbers to at least 6sf) NB Accept bar notation for dot notation to indicate recurring decimals
	eg $1000x - x = 702.70... - 0.702... = 702$ $(999x = 702)$ **and** $\frac{702}{999} = \frac{26}{37}$ **or** $10000x - 10x = 7027.02... - 7.02... = 7020$ $(9990x = 7020)$ **and** $\frac{7020}{9990} = \frac{26}{37}$ **or** $100000x - 100x = 70270.20... - 70.20 = 70200$ $(99900x = 70200)$ **and** $\frac{70200}{99900} = \frac{26}{37}$	shown		A1	for completion to $\frac{26}{37}$ dep on M1 and must use algebra for this final mark to be awarded **No algebra used gets a maximum of 1 mark**
	Working required				**Total 2 marks**

Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2024 · Question 18

18	eg $1000x = 306.306\ldots$ $x = 0.306\ldots$ OR eg $1000\,000x = 306\,306.(\ldots)$ $1000x = 306.306\ldots$		2	M1	M1 for two correct **algebraic equations** involving recurring decimals that when subtracted give a whole number or terminating decimal (306 or 306 000 etc) with intention to subtract. eg $1000x = 306.306\ldots$ and $x = 0.306\ldots$ **or** $1000\,000x = 306306.(\ldots)$ and $1000x = 306.306$ (if recurring dots not shown in both numbers then showing at least **one** of the numbers to at least 6sf)
	eg $1000x - x = 306.306\ldots - 0.306306\ldots = 306$ **and** $\frac{306}{999} = \frac{34}{111}$ or $999x = 306$ **and** $\frac{306}{999} = \frac{34}{111}$ OR eg $1000\,000x - 1000x = 306306.(\ldots) - 306.306 = 306\,000$ **and** $\frac{306\,000}{999\,000} = \frac{34}{111}$ or $999\,000x = 306\,000$ **and** $\frac{306\,000}{999\,000} = \frac{34}{111}$	shown		A1	for completion to $\frac{34}{111}$ dep on M1
	Working required				**Total 2 marks**

Question 12

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2026 · Question 8

Question		Working	Answer	Mark	Notes
*	(a)		$\frac{3}{5}$	1	B1 cao
	(b)		20	1	B1 cao
	(c)		0.04	1	B1
					Total 3 marks

Question 13

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2023 · Question 1

imply a correct method

Q	Working	Answer	Mark	Notes
1	eg $\frac{14}{3}$ **and** $\frac{11}{6}$		3	M1 for both mixed numbers expressed as improper fractions
	eg $\frac{14}{3} \times \frac{6}{11}$ **or** $\frac{28}{6} \div \frac{11}{6}$ **or** $\frac{28n}{6n} \div \frac{11n}{6n}$			M1 seeing this stage gains M2
	eg $\frac{14}{3} \times \frac{6}{11} = \frac{84}{33} = \frac{28}{11} = 2\frac{6}{11}$ **or** $\frac{14}{3} \times \frac{6}{11} = \frac{84}{33} = 2\frac{18}{33} = 2\frac{6}{11}$ **or** $\frac{14}{\cancel{3}^{1}} \times \frac{\cancel{6}^{2}}{11} = \frac{28}{11} = 2\frac{6}{11}$ **or** $\frac{14}{3} \div \frac{11}{6} = \frac{28}{6} \div \frac{11}{6} = \frac{28}{11} = 2\frac{6}{11}$ **or** correct working to $\frac{28}{11}$ **and** writing $2\frac{6}{11} = \frac{28}{11}$ *Working required*	Shown		A1 dep on M2 for conclusion to $2\frac{6}{11}$ from correct working – either sight of result of multiplication eg $\frac{84}{33}$ must be seen or correct cancelling to $\frac{28}{11}$ or complete method using division and common denominators
				Total 3 marks

Question 14

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2023 · Question 1

taken to imply a correct method.

Question	Working	Answer	Mark	Notes
1	$\frac{26}{7}, \frac{13}{8}$ oe		3	M1 both fractions expressed as improper fractions, no need for ÷ or × may be equivalent to those given eg $\frac{52}{14}, \frac{26}{16}$ etc. A student could invert $\frac{13}{8}$ and show multiplication - as shown in the 2nd M1, this mark is then implied.
	$\frac{26}{7}\times\frac{8}{13}$ oe or eg $\frac{208}{56}\div\frac{91}{56}$			M1 **or** for both fractions expressed as equivalent fractions with denominators that are a common multiple of 7 and 8 eg $\frac{208}{56}\div\frac{91}{56}$
	eg $\frac{26}{7}\times\frac{8}{13}=\frac{208}{91}=\frac{16}{7}=2\frac{2}{7}$ **or** $\frac{26}{7}\times\frac{8}{13}=\frac{208}{91}=2\frac{26}{91}=2\frac{2}{7}$ **or** $\frac{\cancel{26}^{2}}{7}\times\frac{8}{\cancel{13}^{1}}=\frac{16}{7}=2\frac{2}{7}$ **or** $\frac{208}{56}\div\frac{91}{56}=\frac{208}{91}=\frac{16}{7}=2\frac{2}{7}$ **or** correct working to $\frac{16}{7}$ **and** writing $2\frac{2}{7}=\frac{16}{7}$ (usually on the first line of working) *working required*	shown		A1 dep on M2 **NB: use of decimals scores no marks (unless used as a check)**
				Total 3 marks

Question 15

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · January 2021 · Question 1

1	e.g. $\frac{16}{5}$ **and** $\frac{11}{6}$ or $\frac{96}{30}$ **and** $\frac{55}{30}$		3	M1 for two correct improper fractions
	e.g. $\frac{\cancel{16}^{8}}{5}\times\frac{11}{\cancel{6}^{3}}$ **or** $\frac{176}{30}$ **or** $\frac{5280}{900}$ oe			M1 correct cancelling or multiplication of numerators and denominators without cancelling
	e.g. $\frac{16}{5}\times\frac{11}{6}=\frac{176}{30}=\frac{88}{15}=5\frac{13}{15}$ **or** $\frac{16}{5}\times\frac{11}{6}=\frac{176}{30}=5\frac{26}{30}=5\frac{13}{15}$ **or** $\frac{\cancel{16}^{8}}{5}\times\frac{11}{\cancel{6}^{3}}=\frac{88}{15}=5\frac{13}{15}$ **or** $\frac{96}{30}\times\frac{55}{30}=\frac{5280}{900}=\frac{88}{15}=5\frac{13}{15}$ NB: a student can show initially that $5\frac{13}{15}=\frac{88}{15}$ and they need to show that LHS = $\frac{88}{15}$	shown		A1 Dep on M2 for conclusion to $5\frac{13}{15}$ from correct working – either sight of the result of the multiplication e.g. $\frac{176}{30}$ must be seen and equated to $\frac{88}{15}$ or $5\frac{26}{30}$ **or** correct cancelling prior to the multiplication to $\frac{88}{15}$ NB: use of decimals scores no marks
				Total 3 marks

Art of Maths Hub International Community Schools

Question 16

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2022 · Question 2

2	eg $\frac{27}{4}$ **and** $\frac{18}{7}$		3	M1	Both fractions expressed as improper fractions.
	$\frac{27}{4}\times\frac{7}{18}$ oe **or** eg $\frac{189}{28}\div\frac{72}{28}$			M1	Invert 2nd fraction or for both fractions expressed as equivalent fractions with denominators that are a common multiple of 4 and 7 (seeing this stage gains M2)
	eg $\frac{27}{4}\times\frac{7}{18}=\frac{189}{72}=\frac{21}{8}=2\frac{5}{8}$ **or** $\frac{27}{4}\times\frac{7}{18}=\frac{189}{72}=2\frac{45}{72}=2\frac{5}{8}$ **or** $\frac{\cancel{27}^{3}}{4}\times\frac{7}{\cancel{18}^{2}}=\frac{21}{8}=2\frac{5}{8}$ **or** $\frac{189}{28}\div\frac{72}{28}=\frac{189}{72}=2\frac{45}{72}=2\frac{5}{8}$ oe if the student clearly shows $2\frac{5}{8}=\frac{21}{8}$ then they only need to complete the LHS to $\frac{21}{8}$ (often done in 1st line of working)	shown		A1	dep M2 conclusion to $2\frac{5}{8}$ from correct working – either sight of the result of the multiplication e.g. $\frac{189}{72}$ must be seen then cancelled or correct cancelling prior to the multiplication with $\frac{21}{8}$ seen. NB entire solution using decimals scores no marks.
					Total 3 marks

Question 17

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2021 · Question 2

Question	Working	Answer	Mark	Notes
2	eg $\frac{18}{7}$ **and** $\frac{9}{8}$ oe		3	M1 both fractions expressed as improper fractions, no need for ÷ or × may be equivalent to those given eg $\frac{36}{14}$ or $\frac{27}{24}$ etc. A student could invert $\frac{9}{8}$ and show multiplication - as shown in the 2nd M1, this mark is then implied.
	eg $\frac{18}{7}\times\frac{8}{9}$ oe **or** oe $\frac{144}{56}\div\frac{63}{56}$			M1 **or** for both fractions expressed as equivalent fractions with denominators that are a common multiple of 7 and 8 eg $\frac{144}{56}\div\frac{63}{56}$
	eg $\frac{18}{7}\times\frac{8}{9}=\frac{144}{63}=\frac{16}{7}=2\frac{2}{7}$ **or** $\frac{18}{7}\times\frac{8}{9}=\frac{144}{63}=2\frac{18}{63}=2\frac{2}{7}$ **or** $\frac{\cancel{18}^{2}}{7}\times\frac{8}{\cancel{9}^{1}}=\frac{16}{7}=2\frac{2}{7}$ **or** $\frac{18}{7}\div\frac{9}{8}=\frac{144}{56}\div\frac{63}{56}=\frac{144}{63}=\frac{16}{7}=2\frac{2}{7}$ **or** correct working to $\frac{16}{7}$ **and** writing $2\frac{2}{7}=\frac{16}{7}$	shown		A1 Dep on M2 for conclusion to $2\frac{2}{7}$ from correct working – either sight of the result of the multiplication or division eg $\frac{144}{63}$ must be seen and then cancelled or correct cancelling prior to the multiplication to $\frac{16}{7}$ **or** writing $2\frac{2}{7}=\frac{16}{7}$ (maybe on first line of working) and correct working as far as LHS $=\frac{16}{7}$ **NB: use of decimals scores no marks**
				Total 3 marks

Question 18

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2024 · Question 4

4	$\frac{18}{7}$, $\frac{28}{9}$		3	M1	for correct improper fractions
	$\frac{18^2}{7^1}\times\frac{28^4}{9^1}$ or $\frac{18}{7}\times\frac{28}{9}=\frac{504}{63}$ oe eg $\frac{18^2}{7}\times\frac{28}{9^1}=\frac{56}{7}$ or $\left(\frac{18}{7}\times\frac{28}{9}=\right)\frac{162}{63}\times\frac{196}{63}=\frac{31752}{3969}$ oe			M1dep	for cancelling fractions fully or cancelling fractions partially and clear intention to multiply (allow arithmetic error in multiplication) or not cancelling and clear intention to multiply (allow arithmetic error in multiplication)
	eg $\frac{18^2}{7^1}\times\frac{28^4}{9^1}=8$ or $\frac{18^2}{7^1}\times\frac{28^4}{9^1}=2\times4=8$ eg $\frac{18}{7}\times\frac{28}{9}=\frac{504}{63}=8$ oe or eg $\left(\frac{18}{7}\times\frac{28}{9}=\right)\frac{162}{63}\times\frac{196}{63}=\frac{31752}{3969}\left(=\frac{8}{1}\right)=8$ *working required*	Shown		A1	Dep on M2 for a correct answer of 8 from fully correct working Candidates may show $8=\frac{8}{1}$ (maybe under the given 8) and then they need only show the given fraction comes to $\frac{8}{1}$
					Total 3 marks

Question 19

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2022 · Question 4

4	$\frac{16}{3}(-)\frac{20}{7}$ or $(5)\frac{7}{21}(-)(2)\frac{18}{21}$		3	M1	for correct improper fractions or fractional part of numbers written correctly over a common denominator (no need for minus sign)
	$\frac{112}{21}-\frac{60}{21}$ or $5\frac{7}{21}-2\frac{18}{21}=3-\frac{11}{21}$ oe or $5\frac{7}{21}-2\frac{18}{21}=4\frac{28}{21}-2\frac{18}{21}$			M1	for correct fractions with a common denominator with minus sign or mixed numbers to the stage shown
	$\frac{112}{21}-\frac{60}{21}=\frac{52}{21}=2\frac{10}{21}$ oe or $3-\frac{11}{21}=2\frac{10}{21}$ or $5\frac{7}{21}-2\frac{18}{21}=4\frac{28}{21}-2\frac{18}{21}=2\frac{10}{21}$	Shown		A1	Dep on M2 for a correct answer from fully correct working If all 3 fractions turned into improper fractions on the first line $\frac{16}{3}-\frac{20}{7}=\frac{52}{21}$ then the student **clearly** needs to show that the LHS $=\frac{52}{21}$
					Total 3 marks

CHALLENGE QUESTIONS

A-Level Maths International Community Schools

Question 20 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2026 · Question 17*

Q	Working	Answer	Mark	Notes
	eg $\frac{14}{5}$ and $\frac{24}{7}$ oe eg $\frac{98}{35}$ and $\frac{120}{35}$		3	M1 for $2\frac{4}{5}$ and $3\frac{3}{7}$ expressed as improper fractions
	eg $\frac{\cancel{14}^{2}}{5}\times\frac{24}{\cancel{7}^{1}}$ **OR** $\frac{336}{35}$ oe eg $\frac{11760}{1225}$			M1 correct cancelling or multiplication of numerators and denominators without cancelling NB: correct cancellation may be implied by subsequent working (eg $\frac{14}{5}\times\frac{24}{7}=\frac{2}{5}\times 24$) – the cancelled factor need not be explicitly crossed out and the resulting 1 may be omitted
	eg $\frac{\cancel{14}^{2}}{5}\times\frac{24}{\cancel{7}^{1}}=\frac{48}{5}=9\frac{3}{5}$ **or** $\frac{14}{5}\times\frac{24}{7}=\frac{336}{35}=\frac{48}{5}=9\frac{3}{5}$ **or** $\frac{14}{5}\times\frac{24}{7}=\frac{336}{35}=9\frac{21}{35}=9\frac{3}{5}$ **or** $\frac{98}{35}\times\frac{120}{35}=\frac{11760}{1225}=\frac{48}{5}=9\frac{3}{5}$ **or** for correct working to $\frac{48}{5}$ **and** writing $9\frac{3}{5}=\frac{48}{5}$ (may be earlier in working) *Working required*	shown		A1 dep on M2, for conclusion to $9\frac{3}{5}$ from correct working – either sight of the result of the multiplication eg $\frac{336}{35}$ oe must be seen **or** correct cancelling prior to the multiplication to $\frac{48}{5}$ NB: use of decimals scores no marks unless used as a check
				Total 3 marks

A-Maths HQ International Community School

Question 21

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2021 · Question 18

18	E.g. $y = 0.4x....$ **and** $10y = 4.x....$ $(10y - y = 4.x - 0.4$ oe) **or** $10y = 4.x....$ **and** $100y = 4x.x....$ $(100y - 10y = 4x - 4$ oe) **or** $100y = 4x.x....$ **and** $1000y = 4xx.x....$ $(1000y - 100y = 4xx - 4x$ oe)		3	M1 for selecting 2 correct recurring decimal expressions and then a demonstration to subtract (If recurring dots not shown then allow each expression to 1 dp e.g. $y = 0.4x....$ **and** $100y = 4x.x....$ as a pair **and** $100y - y$ or $4x.x.... - 0.4x....$) or an answer of $y = \frac{4x-4}{90}$ oe
	E.g. $9y = 4\frac{x}{10} - \frac{4}{10} = \frac{40+x-4}{10}$ oe or $90y = 40 + x - 4$ oe or $900y = 400 + 10x + x - 40 - x$ oe			M1 for a correct subtraction with correct expressions simplified
	Working required	$\frac{36+x}{90}$		A1 dep on M2 oe
				Total 3 marks

Question 22

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2025 · Question 3

3	eg $1 - \frac{7}{20}\left(=\frac{13}{20}\right)$ or $1 - 0.35$ $(= 0.65)$ oe or $100\% - 35\%$ $(= 65\%)$ or $(9 + 43) \div (20 - 7)$ or $52 \div 13$ or (Y : R & B =) 7 : 13 or $\frac{x}{9+43+x} = \frac{7}{20}$ oe or $\frac{7}{20} \times (9 + 43 + x) = x$ oe or $\frac{y-9-43}{y} = \frac{7}{20}$ oe or $\left(1 - \frac{7}{20}\right) \times y = 9 + 43$ oe or $9 \times 35 + 43 \times 35$ $(= 52 \times 35)$		4	M1 for a correct first step **seen** eg $\frac{13}{20}$ or for any correct equation that would lead to 28 or 80 (x = number of yellow bricks and y = total number of bricks)
	$(x =)$ 28 or $(y =)$ 80 or 1820 or 2800			A1 28 or 80 may be seen as a fraction 1820 is the weight of the red and blue bricks 2800 is the total weight of the bricks
	eg ("80" – 9 – 43) × 35 oe or ("80" – 52) × 35 oe or $80 \times \frac{7}{20} \times 35$ oe or "28" × 35 or "1820" ÷ 13 × 7 oe			M1 If the first step is to form a correct algebraic equation then ft their found value of x or y
	Correct answer scores full marks (unless from obvious incorrect working)	980		A1
				Total 4 marks

A* Maths Hub International Community Schools

Question 23

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2023 · Question 15

15 (a)	eg $x = 0.372....$ **and** $100x = 37.272....$ **or** $10x = 3.72....$ **and** $1000x = 372.72....$		2	M1	For 2 recurring decimals that when subtracted give a whole number or terminating decimal eg $100x = 37.272....$ and $x = 0.372....$ or $1000x = 372.72....$and $10x = 3.72....$ **with intention to subtract.** (At least one of the numbers must be shown with recurring dots or to at least 5 sf) **or** $0.3 + 0.0727\ldots$ **and** eg $y = 0.072....$, $100y = 7.2727....$ with intention to subtract.
	eg $100x - x = 37.272.... - 0.372.... = 36.9$ **and** $\frac{36.9}{99} = \frac{41}{110}$ **or** $1000x - 10x = 372.72.... - 3.72.... = 369$ **and** $\frac{369}{990} = \frac{41}{110}$ **or** $100y - y = 7.2727.... - 0.072.... = 7.2$ **and** $\frac{7.2}{99}$ **and** $\frac{3}{10} + \frac{7.2}{99} = \frac{297+72}{990} = \frac{369}{990} = \frac{41}{110}$ oe *Working required*	Shown		A1	Dep on M1 and **use of algebra** for completion to $\frac{41}{110}$ NB: this is a 'show that' question and requires students to clearly show steps that could be used to change the recurring decimal into the given fraction– some may have slight variations to this mark scheme but if the stages can be clearly followed then marks should be awarded.

	Award method marks in either order			
(b)	eg $\frac{\sqrt{125}+\sqrt{80}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$ or $\frac{\sqrt{375}+\sqrt{240}}{3}$ **or** $\frac{5\sqrt{5}+4\sqrt{5}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$ or $\frac{5\sqrt{15}+4\sqrt{15}}{3}$ or $\frac{9\sqrt{5}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$ or $\frac{9\sqrt{15}}{3}$ or $\frac{9\sqrt{5}}{\sqrt{3}} = \frac{\sqrt{405}}{\sqrt{3}}$ oe **or**		3	M1	For rationalising by multiplying by $\frac{\sqrt{3}}{\sqrt{3}}$ or $\frac{-\sqrt{3}}{-\sqrt{3}}$ **or** for $\frac{9\sqrt{5}}{\sqrt{3}} = \frac{\sqrt{405}}{\sqrt{3}}$
	eg $\frac{\sqrt{125}+\sqrt{80}}{\sqrt{3}} = \frac{5\sqrt{5}+4\sqrt{5}}{(\sqrt{3})}$ or $\frac{\sqrt{375}+\sqrt{240}}{3} = \frac{5\sqrt{15}+4\sqrt{15}}{3}$ (must see $\frac{\sqrt{375}+\sqrt{240}}{3}$ before simplifying)			M1	For simplifying the individual surds – either before rationalisation or after rationalisation (for the given surds, we do not need to see the denominator)
	Working required	$\sqrt{135}$		A1	dep on M2 SCB1 for $\sqrt{135}$ gained with no method marks awarded SCB2 for $\sqrt{135}$ and rationalisation also shown or $\sqrt{135}$ and simplifying the numerator shown
					Total 5 marks

A+ Maths International community schools

Question 24 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · January 2021 · Question 15*

15 a	E.g. $x = 4.57....$ **and** $100x = 457.57....$ **or** $10x = 45.757....$ **and** $1000x = 4575.7....$ **or** $x = 0.57....$ **and** $100x = 57.57....$ **or** $10x = 5.757....$ **and** $1000x = 575.7....$		2	M1	for selecting 2 recurring decimals that when subtracted give a whole number or terminating decimal eg 453 or 4530 etc eg $100x = 457.57....$ and $x = 4.57....$ or $1000x = 4575.7....$and $10x = 45.757....$ with intention to subtract. (If recurring dots not shown then allow $10x = 45.757$, $100x = 457.57$, and $1000x = 4575.7$ to at least 5sf) **or** $4 + 0.5757$ **and** eg $x = 0.57....$, $100x = 57.57....$ with intention to subtract.
	E.g. $100x - x = 457.57.... - 4.57.... = 453$ **and** $\frac{453}{99} = \frac{151}{33}$ or $4\frac{19}{33}$ **or** $1000x - 10x = 4575.7.... - 45.757.... = 4530$ **and** $\frac{4530}{990} = \frac{151}{33}$ or $4\frac{19}{33}$ **or** $100x - x = 57.57.... - 0.57.... = 57$ **and** $\frac{57}{99}$ or $\frac{19}{33}$ (so) $4.\dot{5}\dot{7} = 4\frac{19}{33}$ **or** $1000x - 10x = 575.7.... - 5.757.... = 570$ **and** $\frac{570}{990}$ or $\frac{57}{99}$ or $\frac{19}{33}$ (so) $4.\dot{5}\dot{7} = 4\frac{19}{33}$	Shown		A1	for completion to $\frac{151}{33}$ or $4\frac{19}{33}$
15 b	E.g. $\frac{2}{6-3\sqrt{2}} \times \frac{6+3\sqrt{2}}{6+3\sqrt{2}}$ or $\frac{2}{6-3\sqrt{2}} \times \frac{-6-3\sqrt{2}}{-6-3\sqrt{2}}$		3	M1	for rationalising the denominator by multiplying numerator and denominator by $6+3\sqrt{2}$ (or $-6-3\sqrt{2}$)
	$\frac{12+6\sqrt{2}}{36-18\sqrt{2}+18\sqrt{2}-18}$ or $\frac{12+6\sqrt{2}}{18}$ or $\frac{12+6\sqrt{2}}{6^2-\left(3\sqrt{2}\right)^2}$ or $\frac{12+6\sqrt{2}}{6^2-9\times2}$			M1	(numerator may be expanded or denominator may be 4 terms which need to be all correct)
		$\frac{2+\sqrt{2}}{3}$		A1	or for stating $a = 2$ and $b = 3$ for $\frac{2+\sqrt{2}}{3}$ from correct working dep on M2
					Total 5 marks

A+ Maths Hub International community Schools